

LESSON 127 (1.5) GROUND

Lesson Objective

The purpose of this lesson is to introduce the learner to soft-field operations, short-field operations, steep turns, emergency descents and basic instrument maneuvers

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Chapter 06 of the Airplane Flying Handbook - Takeoffs and Departure Climbs:
 - Short field takeoffs.
 - Soft field takeoffs.
 - Chapter 08 of the Airplane Flying Handbook - Airport Traffic Patterns:
 - Short field landings.
 - Soft field landings.
- During this lesson quiz the student on the following and provide instruction where needed.
- Student questions.

References

- Airplane Flying handbook (FAA-H-8083-3C)
 - Chapter 6: Takeoffs and Departure Climbs
 - Chapter 9: Approaches and Landings
 - Chapter 10: Performance Maneuvers
 - Chapter 18: Emergency Procedures
- Instrument Flying handbook (FAA-H-8083-15B)
 - Chapter 7: Basic Flight Maneuvers using analog/Electronic Flight Display

Instructor Guidance

- Encourage the student to chair-fly all procedures introduced in this lesson to reinforce proper sequence and flow. Chair flying is one of the most underutilized study tools in primary training — make it a specific homework assignment, not just a suggestion.
- Utilize instructional videos and a model airplane where available to demonstrate what these maneuvers look like in three dimensions. Short-field and soft-field concepts — particularly ground effect management during soft-field operations — are significantly easier to grasp visually than through verbal description alone.
- Ensure the student has a solid understanding of ground effect before moving on to soft-field procedures. A student who cannot explain how ground effect reduces induced drag and allows the aircraft to fly at lower airspeeds will struggle to understand why soft-field technique works the way it does.
- For soft-field takeoffs, emphasize that continuous movement is essential — stopping on a soft surface risks the nose gear becoming stuck. The student must understand this operationally, not just procedurally.
- When covering short-field takeoffs, stress the importance of not chasing airspeed during the

V_x climb. Set the pitch attitude first and let the airspeed follow — this is a common error and one of the most important habits to establish before the maneuver is flown.

- For steep turns, connect the load factor discussion directly back to what was covered in Lesson 106 on aerodynamics. Students who made that connection in the aerodynamics lesson will find the steep turn concepts much easier to absorb here. Use the load factor chart and the $45^\circ = 1.4G$ / $60^\circ = 2.0G$ relationship as concrete reference points.
- When introducing instrument maneuvers, reinforce the "attitude + power = performance" framework the student already knows from earlier lessons — the same principle applies under the hood. Make this explicit so the student sees instrument flying as an extension of what they already do visually, not as an entirely new skill set.
- For unusual attitude recoveries, emphasize that the recovery sequence for a nose-low attitude is wings level and reduce power before raising the nose. Students who instinctively pull back first when nose-low risk an accelerated stall or structural overspeed — this must be understood clearly before the maneuver is practiced.

Lesson Content

Soft-Field Takeoff and Climb

- "Takeoffs and climbs from soft fields require the use of operational techniques for getting the airplane airborne as quickly as possible to eliminate the drag caused by tall grass, soft sand, mud, and snow and may require climbing over an obstacle" (FAA, 2021).
- "The technique makes judicious use of ground effect to reduce landing gear drag and requires an understanding of the airplane's slow speed characteristics and responses" (FAA, 2021).
- Flaps should be set to 25 degrees. This increases the camber of the wing and therefore increases the amount of lift produced.
- Takeoff roll:
 - When entering the runway, pull the yoke all the way aft to reduce the weight on the front wheel.
 - Do not stop on soft field environments.
 - Minimal braking is to be used, as this can result in the nose gear getting stuck in the grass, mud, sand, etc.
 - As full power is applied, full aft stabilator must be kept.
 - As the nose wheel gets off the runway, release the backpressure slightly to help the main gear get off the runway.
 - Once the airplane is fully off the runway, apply forward stabilator pressure to stay in ground effect until V_x or V_y (depending if there is an obstacle).
 - Crosswind correction must be maintained.
- Departure:
 - After rotation, if there is no obstacle, climb out at V_y.
 - If there is an obstacle, climb out at V_x.
 - Announce positive rate/clear of obstacles and retract the flaps to 10 degrees.
 - Announce positive rate and airspeed acceleration and retract the flaps to 0 degrees.
 - Continue the climb at 79 knots

Soft-Field Approach and Landing

- “Landing on fields that are rough or have soft surfaces, such as snow, sand, mud, or tall grass, require unique procedures. When landing on such surfaces, the objective is to touch down as smooth as possible and at the slowest possible landing speed” (FAA, 2021).
- “A pilot must control the airplane in a manner that the wings support the weight of the airplane as long as practical to minimize drag and stresses imposed on the landing gear by the rough or soft surface” (FAA, 2021).
- The airplane must be held into ground effect in order to touch down softly at minimum speed.
- Approach and landing:
 - The approach is identical to a normal approach.
 - At the round out, reduce the power to idle and stay in ground effect.
 - Begin a flare once the runway seems to be coming up to the airplane.
 - If necessary, a slight addition of power during touchdown will aid in softening the touchdown
- Landing roll:
 - Hold sufficient back-stabilator pressure to keep the nosewheel off the ground until it can no longer aerodynamically be held off the field surface.
 - Use minimal braking.

Short-Field Takeoff and Maximum Performance Climb

- Short runway or obstacle at the end of the runway.
- “To depart from such an area safely, the pilot must exercise positive and precise control of airplane attitude and airspeed so that takeoff and climb performance result in the shortest ground roll and the steepest angle of climb” (FAA, 2021).
- In order to achieve that, V_X must be maintained until clear of the obstacle.
- Re-introduce V_X and explain if needed.
- Flaps should be set to 25 degrees. This increases the camber of the wing and therefore increases the amount of lift produced.
- Takeoff roll:
 - When entering the runway, use the maximum amount of runway available.
 - Once lined up with the centerline, stop on the runway, hold the brakes, and apply full power.
 - Release the brakes once the engine is at full static RPM.
 - Crosswind correction must be maintained.
 - Rotation is at 55 knots.
- Departure:
 - Once off of the runway, pitch for 63 knots until clear of the obstacle.
 - Emphasize to the student to not chase the airspeed. Instead, set a pitch attitude that will get the airplane to a particular airspeed.
 - Announce clear of obstacle.
 - Announce positive rate and retract the flaps to 10 degrees.
 - Announce positive rate and airspeed acceleration and retract the flaps to 0 degrees.
 - Continue the climb at 79 knots.

Short-Field Approach and Landing

- “Short-field approaches and landings require the use of procedures for approaches and landings at fields with a relatively short landing area or where an approach is made over obstacles that limit the available landing area” (FAA, 2021).
- Precise and positive control of the rate of descent and airspeed is required in order to achieve a successful short field landing.
- The landing must result in little floating in order to stop the airplane in the shortest distance.
- Approach and landing:
 - A higher than normal approach is required in order to clear the obstacle on the departure end of the runway.
 - Adjust power to maintain a steeper than normal angle of descent.
 - TRIM!
 - Short field approach speed: 63 knots (1400-1500 RPM).
 - Short final speed: 55 knots (1.2V_{SO}).
 - At the round out, continue the flare to touchdown on the main wheels first in approximately the pitch attitude that will result in a power-off stall just as the power is reduced to idle.
- Landing roll:
 - Upon touchdown, hold this pitch attitude as long as the stabilator remains effective to provide aerodynamic braking, as appropriate.
 - Brakes should be applied evenly and firmly to minimize the after landing roll without skidding.

Steep Turns

- The purpose of the Steep Turns maneuver is to develop a pilot’s smoothness, coordination, orientation, division of attention, and control accuracy while using both outside visual references and flight instruments. The maneuver demonstrates understanding of load factor, coordinated flight, overbanking tendency, and the relationship between bank angle, lift, and stall speed. It reinforces precise control of altitude, airspeed, and bank angle while maintaining situational awareness.
- Procedure: Review Maneuvers Manual
- Talk about the following fundamental concepts:
 - Load Factor
 - As bank angle increases, total lift must increase to maintain altitude.
 - The required increase in lift results in a load factor greater than 1 G.
 - At 45° of bank → 1.4 G
 - At 60° of bank → 2.0 G
 - This additional load increases stall speed, structural stress, and control sensitivity.
 - Pilots must apply increased back pressure to maintain altitude, but avoid excessive input that can lead to an accelerated stall.
- Overbanking Tendency
 - As bank steepens, the outer wing travels faster than the inner wing, generating more lift.

- This causes the aircraft to continue rolling into a steeper bank — known as overbanking tendency.
- The pilot must apply opposite aileron pressure to counteract this tendency and hold a constant bank angle.
- Lift components in a bank.
 - In level flight, total lift acts vertically.
 - In a turn, lift is divided into vertical and horizontal components:
 - Vertical component opposes weight.
 - Horizontal component provides centripetal force to turn the airplane.
 - As bank angle increases, the vertical component decreases, requiring more total lift (increased back pressure) to maintain altitude
- Coordination
 - Steep turns require simultaneous and proportional use of aileron, rudder, and elevator.
 - Improper coordination can cause:
 - Slips (not enough rudder) → altitude loss.
 - Skids (too much rudder) → potential overbanking and stall risk.
 - Coordination keeps the airplane balanced, ensuring smooth heading control and consistent airspeed.
- Pitch, Power & Bank relationship
 - Pitch controls altitude — more back pressure = more lift.
 - Power may need to be slightly increased ($\approx 100\text{--}200$ RPM in a Warrior) to overcome increased drag from higher load factor.
 - Bank determines turn rate and radius — tighter turns require steeper banks, which demand higher load factors and more lift.
 - Proper integration of pitch, power, and bank is key to maintaining a constant altitude and airspeed.
- Altitude, Heading & Rollout
 - Maintain altitude within ± 100 ft, airspeed within ± 10 knots, and bank angle within $\pm 5^\circ$.
 - Lead rollout by $\frac{1}{2}$ the bank angle in degrees (e.g., $20\text{--}25^\circ$ prior to desired heading).
 - Apply coordinated opposite aileron and rudder during rollout to prevent overcorrection.
- Situational Awareness
 - Look outside primarily at the horizon reference point on the cowling relative to the horizon.
 - Divide attention between outside references and instrument cross-check (altimeter, VSI, airspeed).
 - Smooth, continuous control inputs show mastery of aircraft handling under varying load conditions.

Emergency Descent

- To lose altitude rapidly and safely in response to an in-flight emergency, such as smoke or fire, loss of pressurization (in other aircraft types), or the need to quickly reach a lower altitude for medical or environmental reasons. The maneuver trains the pilot to descend promptly while

maintaining positive aircraft control, observing airspeed and load factor limitations, and preparing for a potential landing or further troubleshooting.

- Procedure
 - Review Maneuvers Manual
- Talk about the following fundamental concepts
 - Aerodynamics of descent
 - An emergency descent involves placing the airplane in a steep, controlled descent at a safe airspeed.
 - A bank of approximately 30 to 45 degrees helps increase drag and prevents excessive airspeed buildup.
 - The nose-down attitude decreases lift and increases the rate of descent. The pilot must balance pitch and bank to achieve a rapid descent without exceeding structural limits.
- Load Factor
 - As the aircraft banks and descends, a positive load factor is produced due to increased back pressure required to maintain a controlled path. Typical load factors range from 1.2 to 2.0 G.
 - Maintaining a positive load factor ensures the airplane remains in a stable, loaded condition rather than free fall.
 - Excessive back pressure should be avoided to prevent exceeding load limits or inducing an accelerated stall.
- Airspeed Control
 - The pilot must maintain airspeed at 95 kts. Airspeed should be monitored continuously to prevent flap overspeed (V_{fe}).

Straight-and-Level Flight (Instrument)

- Establish and maintain constant heading and altitude using attitude + power = performance.
- The attitude indicator is primary for pitch and primary for bank when the aircraft is stabilized.
- Pitch control:
 - Adjust pitch to maintain constant altitude (small, smooth yoke movements).
 - Reference the altimeter and VSI to confirm minor trends.
 - Avoid chasing the VSI — use it as a trend instrument, not a control reference.
- Bank control:
 - Use the heading indicator to maintain heading.
 - Cross-check the turn coordinator or ball to ensure coordination and detect small banks early.
 - Make small aileron and rudder corrections together to maintain wings level.
- Power control:
 - Set and hold the power setting appropriate for the desired airspeed (e.g., 2300–2400 RPM in a Warrior).
 - Use airspeed indicator as the primary power instrument in level flight.

- Trim:
 - Trim off all control pressures once altitude and airspeed stabilize.
 - Proper trim reduces workload and helps maintain precise control.
 - Instrument Cross-Check and Interpretation:
 - Begin with a selective radial scan — attitude indicator as hub; supporting instruments as spokes.
 - Correct for small deviations promptly and smoothly (½-bar width corrections on the attitude indicator).
 - Verify performance by comparing multiple instruments (altimeter, VSI, airspeed).
- Common tendencies:
 - Over-controlling due to fixation on one instrument.
 - Failure to recognize small bank or pitch changes early.
 - Slow power response leading to airspeed/altitude drift.

Constant Airspeed Climbs (Instrument)

- Use attitude + power = performance; pitch controls airspeed, and power controls rate.
- Select a specific pitch attitude on the attitude indicator that yields the desired airspeed (for example, 5°–7° nose-up in a Warrior).
- Apply smooth power to full while maintaining coordinated flight with rudder.
- Cross-check primarily with the attitude indicator, and reference supporting instruments:
 - Airspeed indicator – confirms constant climb speed.
 - VSI and altimeter – confirm climb performance trend.
 - Heading indicator + turn coordinator – ensure wings level.
- Trim off control pressures once the airspeeds stabilizes.
- Anticipate level-off approximately 10% of the rate of climb (for example, 50 feet per 500 fpm).
- Transition by reducing pitch and power smoothly to cruise settings while maintaining constant airspeed.

Constant Airspeed Descents (Instrument)

- Use attitude + power = performance; pitch controls airspeed, and power controls rate.
- Select a specific pitch attitude on the attitude indicator that yields the desired airspeed (for example, 5°–7° nose-down in a Warrior).
- Smoothly reduce power while maintaining coordinated flight with rudder.
- Cross-check primarily with the attitude indicator, and reference supporting instruments:
 - Airspeed indicator – confirms constant descent speed.
 - VSI and altimeter – confirm descent performance trend.
 - Heading indicator + turn coordinator – ensure wings level.
- Trim off control pressures once the airspeed stabilizes.
- Anticipate level-off approximately 10% of the rate of descent (for example, 50 feet per 500 fpm).
- Transition by reducing pitch and power smoothly to cruise settings while maintaining constant airspeed.

Turns to Headings (Instrument)

- Establish desired bank angle (normally 15°–20° for instrument training).
- Use the attitude indicator to set and hold bank angle; turn coordinator confirms standard-rate turn.
- Primary instruments:
 - Heading indicator – primary for bank when stabilized.
 - Attitude indicator – establishes bank attitude.
 - Altimeter + VSI – monitor altitude; adjust pitch to maintain level flight.
- Lead the rollout by ½ the bank angle in degrees (e.g., 10° early from a 20° bank).
- Maintain constant power and coordinated rudder throughout.
- Cross-check scan: Attitude → Heading → Altimeter → Turn Coordinator → Repeat.

Recovery from Unusual Flight Attitudes (Instrument)

- Loss of control in-flight (LOC-I): a significant deviation of an aircraft from an intended flight path usually resulting in an airplane upset.
 - This can occur in any phase of flight.
 - The FAA defines an upset as “an event that unintentionally exceeds the parameters normally experienced in flight or training” (FAA, 2021).
 - Pitch attitude greater than 25°, nose up.
 - Pitch attitude greater than 10°, nose down.
 - Bank angle greater than 45°.
 - Within the above parameters but flying at airspeeds inappropriate for the conditions.
- Recognize an unusual attitude by instrument interpretation (not “feel”):
 - Nose-high: Increasing pitch, decreasing airspeed, increasing altitude.
 - Nose-low: Decreasing pitch, increasing airspeed, altitude loss, possible turn indication.
- Recovery (Nose-High):
 - Level the wings with coordinated aileron and rudder
 - Lower the nose to prevent stall — relax back pressure smoothly.
 - Apply maximum power.
 - Return to desired attitude and trim.
- Recovery (Nose-Low):
 - Level the wings using coordinated aileron and rudder & reduce power to prevent overspeed
 - Gently raise the nose to the level flight attitude.
 - Re-trim and resume normal scan.
- Emphasize instrument cross-check (attitude → airspeed → altimeter → VSI → heading → turn coordinator).
- Avoid over-controlling; make small, smooth corrections.
- Reinforce the “power + attitude + trim” relationship throughout recovery.

Radio Communications, Navigation Systems/Facilities, and Radar Services (Instrument)

- Radio Communication
 - Use standard phraseology and proper radio technique (listen before transmitting, think before keying).
 - Ensure clear, concise, and professional transmissions — include who you're calling, who you are, where you are, and what you want.
 - Acknowledge and read back critical instructions (altitude, heading, altimeter settings, runway assignments).
 - Maintain situational awareness by monitoring guard (121.5) or ATIS/CTAF as appropriate.
 - Always verify frequency changes; anticipate upcoming ATC handoffs or frequency zones.
 - Discuss radio calls between the pilot and ATC when inadvertently entering IMC conditions.
- Navigation Systems/Facilities
 - Discuss how GPS can be used (nearest page) after encountering IMC conditions
 - Discuss how instrument approaches work (surface level)
- Radar Services
 - Understand primary radar (range and azimuth only) and secondary radar (transponder-based — Mode A/C/S).
 - Transponder codes:
 - 1200 – VFR, 7500 – Hijack, 7600 – Radio Failure, 7700 – Emergency.
 - When working with ATC radar:
 - Listen for assigned discrete transponder code.
 - Respond to radar vectors using standard-rate turns and precise altitude control.
 - Read back headings and altitudes to ensure clarity.

Post-Brief

- Student self-critique.
- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 128.
- Assign homework:
 - Review unusual attitude recovery procedures and be able to describe the correct recovery sequence for both nose-high and nose-low scenarios
 - Review AFH Chapter 6 (Takeoffs and Departure Climbs) and Chapter 9 (Approaches and Landings) with a focus on short and soft-field procedures.
 - Review AFH Chapter 10 (Performance Maneuvers) — steep turns procedure and common errors.
 - Review the Warrior Maneuvers Manual for steep turns and emergency descent procedures.

Completion Standards

This lesson is complete when the applicant can:

1. Understand the concepts behind the maneuvers introduced.
2. Understand how to perform the maneuvers introduced.